

Optimal Reclamation Bond Contracts under Hidden Costs and Effort: A Numerical Approach to Pipeline Risk

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Abstract

This paper examines how financial assurance mechanisms can be structured to improve pipeline safety in settings with incomplete information. Motivated by repeated regulatory failures and costly environmental damages borne by landowners, it develops a contract-theoretic framework where reclamation bonds incentivize private precaution. The model captures a setting in which pipeline operators privately know their costs of implementing safety measures and control unobservable effort that reduces the risk of a leak. By offering a menu of bond contracts, the state induces operators to self-select by cost type and choose effort levels that balance private costs against the expected loss from forfeiture. Numerical simulations, calibrated to regulatory benchmarks and empirical spill costs in the United States, demonstrate that differentiated bonds effectively sort operators and drive higher safety effort across both low- and high-cost firms. This approach proves robust under a range of economic parameters, maintaining strong incentives for precaution and ensuring participation. The analysis also highlights potential extensions, including adaptive re-bonding tied to incident histories, offering practical tools for regulators to address risks that traditional monitoring often fails to mitigate.

Introduction

In the dead of winter 2015, the frozen surface of the Yellowstone River masked a looming environmental nightmare. Beneath the ice, Bridger Pipeline's Poplar Pipeline ruptured, releasing between 300 and 1,200 barrels of crude oil into the river near Glendive, Montana (Reuters, 2015). Investigators later revealed that aging welds and riverbed erosion had exposed the pipeline to serious structural risks. It was buried at a depth of only eight feet, a threshold that had become dangerously shallow due to seasonal scouring (Inside Climate News, 2015). The incident sparked renewed alarm over thousands of river crossings nationwide, many of which remain vulnerable to unseen degradation. Cleanup efforts were hampered by freezing temperatures, which trapped oil beneath the surface and extended the environmental damage. This stood as a clear example of how infrastructure failure, hidden from sight, can rapidly escalate into large-scale ecological harm.

In December 2022, another catastrophic spill unfolded in rural Kansas. A major rupture in TC Energy's Keystone pipeline discharged approximately 14,000 barrels of crude oil into a tributary of Mill Creek, making it the largest onshore spill in nearly a decade (Kansas Reflector, 2022). Emergency responders quickly mobilized with containment booms and vacuum systems, but the damage had already been done. Hundreds of fish and several mammals were lost as the oil contaminated the surrounding ecosystem (The Guardian, 2022). It was later revealed that a faulty girth weld, identified as vulnerable years prior, had deteriorated under internal pressure.

However, the pipeline has experienced other major spills across multiple states over the years, including a release of 6,592 barrels in South Dakota in 2017 and 4,515 barrels in North Dakota in 2019, alongside numerous smaller incidents, with causes linked to design flaws, manufacturing defects, and regulatory oversight failures (Reuters, 2025). Despite this extensive history, federal

enforcement actions have been limited in scope and largely ineffective. Regulatory agencies have consistently failed to impose penalties or mandate changes that could have prevented future incidents. Since its inception, the Keystone system has recorded more than 20 spills with total damages exceeding \$100 million, yet TC Energy has paid just over \$300,000 in penalties (Kansas Reflector, 2022). This gap between environmental harm and financial accountability reflects a broader failure of federal oversight to enforce meaningful corrective measures.

These cases are not isolated accidents. They reflect a broader regulatory failure in how financial responsibility and safety are managed for hazardous infrastructure. Pipeline operators often struggle to identify, or choose not to disclose, the root causes of leaks. Even under legislative scrutiny, operators have restricted access to data and limited transparency about pipeline integrity (The Guardian, 2022). These stories reveal a persistent problem in pipeline governance. Environmental risk remains underestimated, while regulatory tools fall short in generating strong incentives for precautionary behavior. Crucially, both cases illustrate how asymmetric information allows pipeline owners to underinvest in safety when monitoring is limited or ineffective.

These incidents motivate the central question of this study: how can policymakers promote pipeline safety when operator effort is hidden, and enforcement is imperfect? Spills generate both public harms, such as ecological degradation and lost environmental services, and private harms borne disproportionately by affected landowners who often lack the means to seek redress. While public interventions like inspections and cleanup mandates can address broad societal losses, they often fail to deter negligence *ex ante*. Inspections are sporadic and prone to missing latent risks, and fines are imposed only after damage has occurred, often too late to change firm behavior or fully compensate victims. This research proposes a contract-theoretic model of reclamation bonds as a self-enforcing solution. By requiring firms to post financial guarantees that are forfeited in the

event of a leak, these instruments embed the cost of failure into the operator's private calculus. Unlike retrospective penalties, bonds act *ex ante*, aligning incentives even in the absence of reliable monitoring (Miceli, 2009). They serve a dual purpose by deterring negligence through financial exposure and by ensuring that cleanup funds are available when needed (USDA Forest Service, 2004). As currently implemented in federal surface-disturbance regulations, reclamation bonds are intended to cover site-specific environmental liabilities and promote compliance in settings where *ex post* enforcement may be weak (U.S. Department of the Interior Bureau of Land Management, 2017). The regulatory and informational gaps described above thus motivate the design of a bond menu that screens firms by cost type and elicits precaution through incentive-compatible contracts.

Literature Review

The theoretical framework guiding this study draws from the seminal literature on optimal regulatory design under asymmetric information, where both agent types and their unobservable actions distort traditional enforcement mechanisms. Baron and Myerson (1982) provide the foundational mechanism design solution by constructing self-selecting contract menus that elicit private cost types. Building on this, Baron (1985) applies the model to environmental regulation, demonstrating that uniform policy instruments like pollution taxes fail when regulators lack type information, and that second-best contracts must internalize heterogeneity to maintain efficiency. These principles directly inform the structure of the reclamation bond menu used in this study, where the State incentivizes self-selection based on the agent's private cost of safety effort. Similarly, Grossman and Hart (1992) formalize the core trade-off in moral hazard settings between risk-sharing and incentive provision when actions affect stochastic outcomes, highlighting how unobservable effort is disciplined through financial exposure, as in this model. This idea of a forfeited bond closely resembles how deductibles work in insurance because the bond is only lost if something goes wrong, which gives operators a strong reason to prevent accidents and aligns their interests with public safety (Hirshleifer and Riley 1992).

Extending these insights to the environmental policy domain, Dasgupta, Hammond, and Maskin (1980) argue that optimal pollution control under incomplete information must rely on incentive-compatible mechanisms rather than direct observability, reinforcing the use of financial guarantees over input monitoring. Spulber (1988) formalizes environmental regulation under adverse selection, concluding that second-best contracts must tolerate informational rents, a feature preserved in this study where high-cost types exert lower effort but receive contracts that still satisfy participation. Lewis (1996) expands this framework to settings with dual asymmetry,

involving hidden cost and benefit valuations, and proposes that outcome-contingent nonlinear contracts outperform flat instruments. This notion is echoed in Thomas (1995), who shows that pollution regulation under hidden type and effort requires a hybrid of fixed payments and contingent penalties, which parallels the upfront commitment and outcome-based forfeiture used here.

This contract-theoretic logic is further refined in applied agri-environmental contexts, which offer a parallel to the pipeline sector in both risk structure and enforcement difficulty. Bontems and Thomas (2006) model farmers with private environmental risks and hidden effort, showing that optimal contracts must trade off distortion and incentive alignment. Peterson and Boisvert (2004) similarly incorporate heterogeneity in both technology and risk preferences, concluding that incentive-compatible contracts must combine fixed and outcome-dependent payments. Moxey, White, and Ozanne (1999) compare uniform and differentiated contract menus and find that menus sensitive to heterogeneity improve efficiency, which justifies the use of a bond menu where agents reveal their types. Wu and Babcock (1996) also emphasize the importance of designing screening contracts that admit informational rents to induce truthful selection, a property reflected in this study's bond structure, where high-cost types retain expected surplus.

Several studies underscore the insurance-effort tradeoff in environmental schemes. Fraser (2002) demonstrates that excessive insurance under risk aversion can suppress desirable effort levels, emphasizing the need for contingent instruments that reward actual outcomes. Rollins and Briggs (1996) arrive at a similar conclusion in wildlife compensation, showing that full indemnification encourages moral hazard and that payments should be conditioned on verifiable outcomes. Miceli (2009), writing from a law-and-economics perspective, frames breach penalties as optimal when enforcement is incomplete, supporting this model's interpretation of bond forfeiture as an implicit

damage payment. These contributions bolster the rationale for using a bond that is refunded or forfeited based solely on observed leaks rather than unverifiable inputs.

When direct monitoring of effort is costly or unreliable, the literature emphasizes financial enforcement tools. Choe and Fraser (1998) show that imperfect monitoring necessitates higher incentive payments and raises the overall cost of achieving compliance. Their follow-up study (Choe and Fraser, 1999) models stochastic compliance signals and argues for contracts that rely more on outcome-based design, which is precisely the approach taken in this study. Ozanne, Hogan, and Colman (2001) confirm that as monitoring becomes infeasible, optimal policies shift toward financial penalties that deter noncompliance *ex ante*, not through detection but through built-in risk. Chambers and Quiggin (1996) support this approach by demonstrating that multidimensional, unverifiable effort requires output-based incentives. Similarly, Strausz (1997) finds that delegated monitoring with asymmetric information reduces enforcement effectiveness, reinforcing the value of outcome-contingent tools such as the bond structure analyzed in this paper.

A core contribution of this study lies in addressing dual asymmetry, where both effort and cost type are unobservable to the regulator. Chavas and Holt (1996) model this empirically in a production context, revealing how both technological heterogeneity and behavioral variation alter agent responses, reinforcing the rationale for modeling different cost types. Ferraro (2008) critiques environmental payment schemes that fail to address dual asymmetry, warning that contracts not tailored to hidden traits risk attracting low-effort agents. Hart and Latacz-Lohmann (2005) show that even without hidden types, unobservable effort alone creates substantial efficiency losses in environmental contracting, which are only magnified when type is also private. Sheriff (2009) further explores this, demonstrating that second-best policy must tradeoff between participation and distortion. Smith and Tomasi (1999) reach a similar conclusion in a setting with

multiple agents, showing that nonlinear instruments can approximate full information outcomes under incentive compatibility, much like the dual bond structure discussed in this thesis.

The closest structural parallel to this study is Mason and Plantinga (2013), who model carbon contracts under dual asymmetry and solve them numerically for optimal menus that balance participation and effort. Their framework informs both the calibration logic, and the numerical method used in this study. Ozanne and White (2003) compare input charges to quotas under hidden information, finding that properly designed financial instruments are sufficient to overcome both adverse selection and moral hazard. These conclusions validate the focus on menu-based contracts that elicit truthful behavior where direct enforcement is weak.. Finally, Hashimoto, Hamada, and Hosoe (2012) offer a numerical solution framework for nonlinear contract problems under asymmetric information. Their simulation-based method justifies this study's computational approach, where the complexity of agent response, incentive constraints, and welfare optimization precludes a closed-form solution.

While the theoretical literature offers strong guidance on designing incentive-compatible contracts, the policy relevance of this study depends on how regulation is structured in practice. In the United States, pipeline oversight involves a layered system of federal, state, and private enforcement tools that combine ex post liability rules with limited ex ante financial safeguards. Understanding this context is essential to motivate the model developed in this paper.

On federal lands, the Bureau of Land Management (BLM) requires performance and reclamation bonds under 43 CFR §2805.20 to ensure site cleanup, project compliance, and environmental restoration. These bonds are calculated using site-specific cost estimates and are periodically reviewed for adequacy. Although these rules traditionally applied to mining and surface-disturbing activities, a 2024 administrative update expanded bonding obligations to pipeline developments as part of a broader move toward financial responsibility for linear infrastructure (Bureau of Land Management, 2024a; 2024b).

Operational regulation, however, is handled separately. The Pipeline and Hazardous Materials Safety Administration oversees design standards, emergency preparedness, and excavation damage prevention through 49 CFR §195.442. While these rules govern behavior and reporting, they do not mandate reclamation bonds or financial reserves. Enforcement occurs primarily through post-violation inspections and civil penalties, which are limited in scope and often imposed after harm has already occurred (PHMSA, 2024).

Some states operate under formal agreements with PHMSA that delegate certain enforcement powers. In addition, several states impose their own bonding or financial assurance rules for pipeline projects. However, the strength and coverage of these mechanisms vary widely. This unevenness, combined with institutional separation between land use approval and operational oversight, creates a fragmented system that often leaves ex ante precaution underenforced.

Beyond bonding and administrative penalties, multiple federal laws impose financial liability after an incident. The Oil Pollution Act of 1990 establishes strict liability for oil spills and created the Oil Spill Liability Trust Fund, which is financed by industry fees and administered by the federal government. The fund pays for cleanup and compensates claimants when responsible parties cannot or do not pay, while retaining authority to recover costs from operators (U.S. Environmental Protection Agency, 2025). Similarly, the Price-Anderson Act provides pooled liability coverage for nuclear incidents, combining private insurance with industry-wide contributions to ensure a compensation mechanism for the public (Congressional Research Service, 2025). These programs function as enterprise-backed public insurance funds, but they operate only after damage has occurred.

Private insurance adds another layer of financial protection. Most pipeline operators hold liability coverage, and insurers conduct independent risk assessments that theoretically align with safety goals. However, insurance contracts are not public, vary in coverage levels, and may generate moral hazard if operators rely on coverage instead of investing in risk reduction. Payouts may also be subject to disputes or delays, limiting their effectiveness as immediate remedies for affected landowners. Negligence rules and tort-based claims offer a final legal channel for compensation. While these can result in financial recovery, they require landowners to pursue litigation and establish fault. This process is often costly, slow, and uncertain, particularly when dealing with large firms that have greater legal resources. These barriers limit the practical deterrence value of ex post liability rules and reduce their protective effect for private parties.

Taken together, this mix of federal liability statutes, delegated state enforcement, insurance, and tort law creates a largely reactive system that activates only after harm has occurred. Reclamation bonds, by contrast, function as forward-looking mechanisms that secure funds and influence operator behavior before a spill. They are not substitutes for existing remedies but a complement that strengthens ex ante precaution and ensures the availability of funds without relying on delayed enforcement. By embedding financial consequences into the initial project approval, bond instruments directly address the weaknesses of fragmented oversight and incomplete deterrence. This framing grounds the contract-theoretic model developed in this paper, where differentiated bonds screen for cost types and elicit private effort in the absence of observable inputs or complete enforcement.

Model

Building on the theoretical foundations and policy context, this section develops a formal model to analyze how reclamation bond design can address the dual problem of unobservable effort and private cost heterogeneity in pipeline safety. The interaction is framed as a principal–agent problem in which the State (principal) seeks to induce safety effort from a Pipeline Owner (agent) despite being unable to observe either the agent’s preventive actions or their internal cost structure. Specifically, the agent chooses a level of safety effort that lowers the probability of a leak but incurs a private cost.

To overcome this information asymmetry, the State offers a menu of bond contracts, each defined by a bond amount that is forfeited in the event of a leak. The design of these contracts relies on self-selection, with each agent expected to choose the contract intended for their type and to exert the corresponding optimal level of effort. This mechanism leverages endogenous effort choices and observable outcomes to align private incentives with public safety goals. The interaction begins with each pipeline operator privately knowing their own cost type, which is either low-cost (c_L) or high-cost (c_H). This type is not known to the state (the principal). A low-cost type can implement pipeline safety measures at a lower internal cost, while these same measures impose significantly higher costs on a high-cost type. This asymmetry of information introduces a moral hazard problem where the state cannot directly observe the agent's cost type, or the amount of effort exerted to prevent pipeline failure.

Following the type assignment, the State designs a menu of contracts, each specifying a bond amount (B) that the agent must post as financial collateral. This bond is forfeited if a pipeline leak occurs. On the other hand, if no leak occurs, the agent recovers the bond in full. The contract does

not directly mandate an effort level; it is structured so that the agent voluntarily chooses an effort level to minimize their own expected loss. Thus, the bond serves as a self-enforcing mechanism that aligns the agent's incentives with the State's safety objectives despite asymmetric information. This overall setting can be characterized as a Stackelberg-type game, where the State acts as the leader and moves first by announcing the menu of contracts, and the agent (follower) then responds by selecting a contract and choosing effort.

After observing the menu, each agent selects the contract intended for their type. The state sets the bond levels carefully to ensure incentive compatibility, where each type prefers its contract over mimicking the other. Once the agent selects a contract, they privately choose an effort level $e \in [0, \infty)$ to reduce the likelihood of a pipeline leak. A higher e implies greater effort and greater safety at a greater cost. In this framework, the effort variable e represents the level of safety investment chosen by the agent, such as pipeline depth, monitoring frequency, or system pressurization, while the cost parameter c_i captures the agent's marginal cost of effort, which varies across firms due to differences in construction technology, terrain conditions, and operational efficiency.

Agent's Problem

Let the agent be risk neutral and let the cost of exerting effort be represented by a quadratic function of the form $c(e, c_i) = c_i e^2$, where $e \geq 0$ denotes the level of safety effort and $c_i \in \{c_L, c_H\}$ reflects the agent's private cost type, which varies across operators. A low-cost type is characterized by c_L , and a high-cost type by c_H , with $c_L < c_H$. This structure implies that low-cost agents can implement safety measures at a lower internal cost. The total effort cost is increasing and convex in effort, consistent with the idea that more intensive safety actions are progressively more expensive.

The probability of a pipeline leak, denoted $q(e)$, declines with higher effort and follows the exponential form $q(e) = q_0 e^{-\alpha e}$, where $q_0 \in (0,1)$ represents the baseline probability of failure when no effort is exerted and $\alpha > 0$ is a sensitivity parameter that governs the effectiveness of effort in reducing risk. This specification satisfies $q'(e) < 0$ and $q''(e) > 0$, capturing the assumption that safety effort has diminishing marginal returns. That is, initial increments in effort substantially reduce leak risk, but additional effort becomes less effective at the margin. In practice, firms may rely on internal monitoring technologies such as pressure sensors and flow measurement systems, among other technologies that signal elevated leak risk, reinforcing the assumption that agents perceive a consistent relationship between safety effort and expected pipeline failure (The Equity Engineering Group, 2024).

Given a contract that specifies a bond amount, the agent's expected utility reflects three components. First, the agent faces the possibility of forfeiting the bond in the event of a pipeline leak, generating an expected loss proportional to the probability of failure. Second, the agent incurs a private internal cost associated with the exertion of safety effort. Third, the agent receives a fixed operational revenue, denoted by R , which represents the value of maintaining an active pipeline permit or the market return from pipeline operations. This revenue is assumed to be constant across types and independent of effort, allowing the analysis to focus on incentive effects rather than output variability. Together, these components yield the agent's expected utility function, which can be expressed in terms of effort, cost type, and bond size.

Hence, the expected utility function of the agent is given by $U(e; c_i, B) = R - c_i e^2 - q(e)B$, where $q(e)B$ is the expected loss due to leak (probability of leak $\times$ bond forfeited).

The agent selects an effort level e to maximize expected utility, conditional on their private cost type and the bond amount specified in the contract. The optimal effort satisfies the first-order

condition obtained by differentiating the utility function with respect to effort, which gives the following equation:

$$U_e = -2c_i e - q'(e)B = 0 \Rightarrow q'(e^*)B = -2c_i e^* \dots\dots\dots (1)$$

This condition characterizes the agent's optimal response by equating the marginal benefit of reducing expected bond forfeiture, given by the left-hand side, with the marginal cost of exerting effort, given by the right-hand side. Since agents with lower cost face a smaller marginal disutility from effort, they will choose a higher effort level for any given bond. This creates a natural screening mechanism in which the bond level indirectly induces type separation through effort choice.

Principal's Problem

The State, assumed to be risk-neutral, faces the problem of designing contracts that induce each pipeline operator to exert safety effort in accordance with the regulator's objectives, despite the presence of private information regarding cost type. To address this, the State offers a menu that consists of two contracts, one intended for agents with a lower cost of effort and the other designed for agents with a higher cost. The contracts are designed to satisfy two key constraints, namely incentive compatibility (IC), which ensures that each agent selects the contract intended for their type, and individual rationality (IR), which guarantees that participation is preferable to opting out.

Let B_L and B_H denote the bond levels offered to agents with cost types c_L and c_H , respectively. Given these bond levels, each agent chooses their effort level $e^*(c_i, B)$ according to the first-order condition derived in equation (1). Accordingly, let $e_L = e^*(B_L, c_L)$ and $e_H = e^*(B_H, c_H)$ denote the equilibrium effort levels exerted by the lower-cost and higher-cost agents, respectively. Moreover,

let $\gamma \in (0, 1)$ represent the fraction of lower-cost agents in the population, so that $1 - \gamma$ denotes the share of higher-cost agents.

When a pipeline leak occurs, the landowner incurs a monetary loss denoted by M , which in this model captures reductions in private wealth, such as land value loss, water contamination, or disruption to productive use¹. In the absence of a leak, the landowner retains their full initial wealth, denoted by λ . The State is assumed to value landowner welfare and therefore seeks to minimize expected social losses arising from pipeline failures and maximize the landowner's utility. The landowner is modeled as risk-averse, with utility defined over wealth. If a leak occurs, his utility is given by $U_L^{\text{leak}} = \sqrt{\lambda - M}$, whereas in the case of no leak, his utility is $U_L^{\text{no leak}} = \sqrt{\lambda}$ and $M \leq \lambda$. The expected utility of the landowner, conditional on the agent's effort level e_i , is therefore:

$$EU_L(e_i) = q(e_i) \sqrt{\lambda - M} + (1 - q(e_i)) \sqrt{\lambda} \quad \dots\dots\dots (2)$$

The objective of the State is to maximize the expected utility of the landowner, aggregated across both types of agents and weighted by their population shares. The resulting objective function is:

$$EU = \gamma EU_L(e_L(B_L)) + (1 - \gamma) EU_L(e_H(B_H)) \quad \dots\dots\dots (3)$$

¹ While pipeline leaks can also generate substantial environmental damages, this model restricts attention to private losses borne by the landowner. This choice reflects the fact that environmental harms are already addressed by legal frameworks such as the Oil Pollution Act of 1990 and the Oil Spill Liability Trust Fund, as discussed in the literature review, which provide dedicated mechanisms for remediation and compensation. By contrast, landowners often lack equivalent protection or bargaining power when seeking compensation for private harms, particularly in cases involving smallholders facing large operators. Focusing on private wealth loss avoids conflating the model with damages that are institutionally internalized elsewhere and highlights the gap in ex ante protection for directly affected property owners.

Where e_L and e_H are the effort levels chosen by agents with cost parameters c_L and c_H , respectively. These effort levels are not directly selected by the State but are implicitly determined through the agents' responses to the bond levels offered in the contract menu. The optimization problem in equation (3) is governed by two key sets of constraints that ensure the contract functions as intended. The incentive compatibility (IC) conditions guarantee that each type of operator selects the contract designed for them, indicating the low-cost agent prefers to choose the contract offering bond B_L and exert effort e_L rather than mimicking the high-cost contract, which is expressed as:

$$R - c_L e_L^2 - q(e_L)B_L \geq R - c_L e_H^2 - q(e_H)B_H, \dots\dots\dots (4)$$

Conversely, the high-cost agent finds it optimal to select their designated contract, formally:

$$R - c_H e_H^2 - q(e_H)B_H \geq R - c_H e_L^2 - q(e_L)B_L, \dots\dots\dots (5)$$

Alongside these, individual rationality (IR) constraints ensure that participation is worthwhile for both types, meaning each agent secures at least a non-negative expected utility by accepting the contract over opting out. This requires the following equations to be satisfied for the low-cost and the high-cost operators, respectively as below:

$$R - c_L e_L^2 - q(e_L)B_L \geq 0 \dots\dots\dots (6)$$

$$R - c_H e_H^2 - q(e_H)B_H \geq 0, \dots\dots\dots (7)$$

Together, these constraints shape the principal's choice of bond levels to induce truthful selection and adequate precaution, aligning private decision-making with broader safety objectives. The State's challenge is to select bond levels such that each agent type chooses the intended contract and responds with the corresponding effort level, leak probabilities are minimized through precaution, landowner utility is maximized, and participation of each agent is ensured.

Among the four constraints presented, only two are typically binding in the optimal contract solution. The incentive compatibility constraint for the agent with lower cost, given by equation (4), is binding because this type has a strategic incentive to mimic the contract intended for the higher-cost type to secure more favorable terms. Likewise, the individual rationality constraint for the higher-cost agent, represented by equation (7), must also bind, as this agent is more difficult to incentivize and must be guaranteed at least a non-negative utility to participate. The remaining two constraints, namely the individual rationality condition for the low-cost agent and the incentive compatibility condition for the high-cost agent, are typically not binding in optimal contract design under standard assumptions of monotonicity and utility ordering. Specifically, if the low-cost agent has a higher marginal return to effort and prefers their designated contract over mimicking the high-cost one, their participation constraint is usually satisfied with slack. Similarly, the high-cost agent typically has no incentive to imitate the low-cost contract due to its higher effort requirement. However, because this model is solved numerically rather than analytically, these assumptions should be verified directly for the calibrated baseline to ensure that they hold in the simulated equilibrium. Combining the objectives and constraints, the state's optimization problem becomes:

$$\max_{B_L, B_H} \quad \gamma EU_L(e_L(B_L)) + (1 - \gamma) EU_L(e_H(B_H))$$

Subject to

1. $q'(e_i)B_i = -2c_i e_i$ for $i \in \{L, H\}$
2. $R - c_L e_L^2 - q(e_L)B_L \geq R - c_L e_H^2 - q(e_H)B_H$
3. $R - c_H e_H^2 - q(e_H)B_H \geq 0$

This formulation constitutes a standard mechanism design problem with moral hazard and heterogeneous types. The bond levels, which are the choice variables in this setup, serve as policy instruments that indirectly influence agent behavior by shaping effort incentives, thereby lowering leak risks and improving expected landowner welfare.

Given the structure of the model, particularly the implicit nature of the agent's optimal effort response and the presence of nonlinear constraints, a closed-form analytical solution is not feasible. As a result, the analysis proceeds using numerical methods where apt simulations are conducted in R to implement the model's equations, calibrate parameter values, and evaluate outcomes under varying bond levels and agent characteristics. The following section presents and interprets the results of the numerical simulations.

Numerical Implementation, Results, and Discussion

In this section, the optimization model is solved using an iterative numerical approach, calibrating parameter values aptly to assess the impact of bond levels on effort, leak risk, and landowner welfare across agent types. To ensure the analysis reflects real-world conditions, the choice of parameters is grounded in empirical studies and regulatory practice. Performance bonding has long been employed as a regulatory safeguard in environmentally sensitive sectors. The U.S. Forest Service (2004) outlines in its Bonding and Reclamation Guide that financial assurance mechanisms must be calibrated to cover full restoration costs in the event of operator noncompliance, while similar standards are enforced under the Office of Surface Mining's Directive 1005 (Office of Surface Mining Reclamation and Enforcement, 2020). However, multiple Government Accountability Office reports (GAO, 2010a, 2010b) have highlighted that outdated bonding levels and insufficient oversight leave taxpayers and landowners vulnerable to cleanup liabilities. Recognizing these deficiencies, the Bureau of Land Management has recently proposed a 15–20-fold increase in minimum bond requirements, citing the growing cost of site remediation (Sauer, 2023). In parallel, NOAA (2025) and Cohen (2010) emphasize that the true economic cost of oil spills goes far beyond direct cleanup expenses, encompassing long-term ecological degradation, lost use values, and reputational harm. These findings provide institutional and economic justification for calibrating the model's bond and damage parameters at levels consistent with actual risk exposure and regulatory expectations.

To operationalize these regulatory insights, this study calibrates its simulation using empirically grounded and economically consistent parameter values accordingly. To represent environmental damages from a pipeline leak, this study uses cleanup cost estimates drawn from the Pipeline and Hazardous Materials Safety Administration, which reports average per-gallon spill response costs

of approximately 200 dollars for crude oil spills, with some cases reaching over 4000 dollars per gallon depending on incident severity, cleanup difficulty, and location sensitivity (PHMSA, 2023). Taking a modest reference case involving 1000 gallons of spilled crude oil, the resulting damage value of 200000 dollars represents a conservative estimate of environmental harm. Cost heterogeneity between agent types is captured by assigning the low-cost operator a marginal effort cost of 1000 dollars for C_L and the high-cost operator a marginal cost of 2000 dollars for C_H , reflecting plausible differences in pipeline technology, construction environments, and operational efficiency. The baseline probability of a leak (q_0) in the absence of any effort is set at 0.05, while the leak to effort decay (α) parameter is calibrated at 0.2 to capture diminishing returns to safety investment. To maintain analytical tractability, the landowner's wealth is set to $\lambda = 1,000,000$, and the damage term (M) is set at \$200,000, implying a 20 percent utility-relevant loss in the event of a leak, and the pipeline operator's revenue is set at $R = \$18,000,000$. This calibration aligns with industry scale, as TC Energy reported quarterly profits exceeding \$912 million from its U.S. natural gas pipeline operations alone (Reuters, 2024), while Targa Resources projected annual core profits surpassing \$4.65 billion (Pipeline & Gas Journal, 2025). A revenue level of \$18 million therefore, reflects a conservative estimate appropriate for modeling a localized or mid-sized project within the broader pipeline industry. The population share of low-cost operators is set at $\gamma = 0.5$ to facilitate symmetric benchmarking across agent types. Together, these parameters provide a realistic foundation for numerically exploring incentive-compatible bonds. Table 1 summarizes the baseline parameter values applied in the numerical simulations in the appendix section.

When the model is numerically optimized using the calibrated parameters described above, the resulting optimal bond levels are $B_L^* = \$150000$ and $B_H^* = \$350000$, with corresponding effort levels of $e_L^* = 0.6576$ units for low-cost agents and $e_H^* = 0.7527$ units for high-cost agents. The

resulting expected utility for the landowner (EU^*) is approximately 995.42. These values are consistent with theoretical expectations. Notably, the optimal bond amounts are consistent with financial assurance standards currently enforced by the Bureau of Land Management, which requires pipeline operators to post a minimum lease bond of \$150,000 and a statewide bond of \$500,000 per operator per state for surface-disturbing activities which reinforces the practical relevance of the model (BLM, n.d.). The lower bond reflects low-cost operators' efficiency, requiring less collateral for precaution. The higher bond for high-cost types compensates for their steeper costs by sharpening incentives to avoid forfeiture. Despite their higher cost, these agents exert more effort than low-cost operators, an intentional feature of the bond structure to preserve incentive compatibility. The higher bond assigned to high-cost operators creates a stronger financial incentive to avoid forfeiture, which offsets their higher marginal cost and results in greater effort. This illustrates how bond differentiation achieves separation and maintains incentive compatibility. The landowner's high expected utility indicates that the contract menu effectively aligns agent behavior with public safety objectives and minimizes leak risk across the operator population, as reflected in the optimal effort levels where the probability of a leak drops to approximately 4.38 percent for low-cost operators and 4.30 percent for high-cost operators from the baseline 5 percent.

Beyond the baseline calibration, six comparative static exercises are presented in Appendix Plots 1 through 6 to examine how variations in key parameters influence the structure of optimal contracts. In Plot 1, changing the damage value M does not affect bond levels or effort because, in this model, the agent's optimization depends only on their own cost of effort, revenue, and the expected loss from bond forfeiture. The damage parameter M enters exclusively into the landowner's utility function and does not appear in the agent's objective or the incentive

constraints. Since the bond is forfeited entirely in the event of a leak, its size is calibrated to satisfy incentive compatibility and individual rationality, both of which are independent of the magnitude of M . As a result, bond and effort levels remain constant across changes in environmental damage. However, the expected utility for the landowner declines with higher M because the utility loss from a leak becomes larger, even though agent behavior remains unchanged.

In Plots 2 and 3, changing the cost parameters c_H and c_L affects the effort levels of the corresponding agent types but does not change the bond levels. This occurs because the agent chooses effort to balance the marginal cost with the marginal benefit of avoiding bond forfeiture. When c_H increases, the high-cost agent finds effort more expensive and reduces effort accordingly, but the low-cost agent faces no change. Similarly, as c_L rises, the low-cost agent also reduces effort while the high-cost agent sees no change in his effort. However, bond levels remain feasible and unchanged as they continue to support type separation. The expected utility for the landowner declines in both cases because lower effort increases the probability of a leak, leading to greater expected loss.

In Plot 4, varying the agent's revenue R does not affect the bond levels, effort choices, or the landowner's expected utility. This result follows directly from the structure of the model, since revenue enters the agent's utility function additively and is independent of both effort and bond. As a result, changes in R shift the agent's total utility but do not affect their marginal incentives to exert effort. Because the incentive compatibility and individual rationality constraints are already satisfied at the calibrated bond levels, increasing R merely raises the agent's surplus without altering their behavior. The State therefore has no reason to adjust the contract, and leak probabilities remain unchanged. Consequently, the landowner's expected utility stays constant across revenue values.

In Plot 5, changing the landowner's wealth λ affects expected utility but leaves bond levels and effort unchanged. This is because the landowner's utility loss from a leak depends on the difference between $\sqrt{\lambda}$ and $\sqrt{\lambda - M}$, so as λ increases, the relative impact of damage becomes smaller, leading to a smaller utility loss and therefore higher expected utility. However, since λ does not enter the agent's utility function or the incentive constraints, it does not influence the agent's behavior or the optimal bond design. As a result, effort and bond levels remain constant, while expected utility rises solely due to the mechanical effect of increased wealth reducing the marginal impact of a leak.

In Plot 6, as the share of low-cost agents γ increases, the landowner's expected utility declines while bond and effort levels remain unchanged. This occurs because in the optimized contract, high-cost agents are assigned a higher bond and exert more effort than low-cost agents to satisfy incentive compatibility. As the population shifts toward more low-cost agents who exert less effort, the average leak probability rises, reducing expected landowner utility. Since the bond levels already support correct self-selection and satisfy all constraints, the State does not adjust them in response to changes in agent composition. Table 2 summarizes the results of the comparative statics, highlighting how changes in key parameters influence bond levels, effort choices, and expected landowner utility.

In Plot 7, increasing the decay rate parameter α causes both high-cost and low-cost firms to increase their effort levels. A higher α means that each unit of effort becomes more effective at reducing the probability of a leak, so the marginal benefit of effort increases. Firms respond by exerting more effort because they can now reduce their expected bond forfeiture more efficiently. As a result, the landowner's expected utility also rises steadily, since the overall probability of a leak falls with greater effort from both types of firms.

In Plot 8, as the base leak probability q_0 increases, both types of firms respond by exerting more effort. A higher q_0 raises the overall likelihood of a leak for any given effort level, which means the expected penalty from forfeiting the bond becomes larger. To avoid this higher expected loss, firms increase their effort to bring the leak probability back down. Even though firms increase effort in response to a higher base leak risk, the total probability of a leak is still worse than before. As a result, the landowner's expected utility declines, because the environment is still riskier overall despite the firms trying harder.

In summary, the results confirm that a menu of reclamation bonds can successfully induce self-selection and precautionary effort under dual asymmetry. Although typically associated with end-of-life restoration, these instruments can also function as hybrid tools that provide financial incentives for ex ante care and offer safeguards against operational risks, including accidental spills. The baseline simulation shows that optimal bond levels differ across agent types, with higher-cost operators receiving stronger financial incentives to ensure participation and separation. Comparative statics further demonstrate that while expected utility for the landowner is sensitive to parameters such as environmental damage, agent cost, landowner wealth, and population composition, the bond levels and effort responses remain stable within feasible ranges, preserving both incentive compatibility and individual rationality. These findings align with the theoretical structure of the model, where the bond mechanism shapes behavior through the logic of expected forfeiture rather than through direct ties to environmental valuation or revenue generation. Given these results, the model offers a structured framework that could inform federal agencies such as the BLM in refining land-based bonding programs to better address both ex-ante care and financial accountability for environmental harm.

Extensions and Future Research

While the current model focuses exclusively on maximizing landowner welfare by reducing leak risk, a natural extension would introduce a convex weighting parameter $\theta \in (0, 1)$ to balance landowner safety with operator welfare. This approach reflects a more general benevolent planner framework and allows the model to capture broader policy objectives. Incorporating both agent surplus and external damages into a single welfare-weighted formulation would provide a more comprehensive lens through which to evaluate regulatory tradeoffs beyond environmental protection alone.

Another extension would incorporate the time value of money into the firm's decision problem. The current framework treats the bond amount as a static financial stake, but in practice, firms discount future payouts relative to present costs. To reflect this, the firm's expected utility could be modified by introducing a continuous discount factor $\delta \in (0, 1)$, where the expected cost of forfeiting a bond after t years becomes $\delta^t B$. Conversely, if the bond is held in a trust accruing interest at rate r , the opportunity cost of capital could be represented by adjusting the firm's objective to include $B(1 + r)^t$, capturing the growth of the foregone return. In this design, if the trust fund is structured so that the growth rate r exactly offsets the discount rate δ , then the present discounted value of the returned funds (bond principal plus interest) equals the original amount posted, ensuring no net financial loss to the firm over time (US EPA, 1996). This design ensures that firms perceive no net financial loss from compliant behavior over time, preserving incentives even when projects are long-lived. These adjustments enable a richer analysis of long-term projects by aligning the timing of bond forfeiture or refund with the firm's opportunity cost of capital, thereby more accurately reflecting the true burden of the up-front investment. In practice, such mechanisms are already reflected in some U.S. state-level financial assurance rules, which use

interest-bearing trust funds and require discounting future liabilities using real interest rates (Nebraska Department of Environment and Energy, 2021).

Future versions of the model could also incorporate risk aversion on the part of pipeline operators, which would influence their perception of bond forfeiture uncertainty, intensify the tradeoff between precaution and perceived risk, and potentially reshape the structure of optimal contracts. In such a setting, the agent's utility function would take the form $U(e; c_i, B) = E[v(R - q(e)B)] - c_i e^2$, where $v(\cdot)$ is a concave function capturing risk aversion and E denotes the expected utility over possible outcomes with and without a leak. Further extensions could also link operator revenue to effort through performance penalties or bonuses, adding another incentive margin. Alternatively, the pipeline company could be modeled as a regulated monopoly that maximizes profit subject to safety constraints, enabling analysis of how bonding requirements or liability rules affect voluntary compliance and investment in risk mitigation.

Another way this model can be expanded is by introducing pipeline operating pressure $p \in [0, \hat{p}]$ as an additional decision variable alongside safety effort, allowing for a richer representation of the tradeoff between production incentives and environmental risk. In this setting, revenue would be defined as $R(p) = P \cdot TP(p)$, where $TP(p) = TP\left(\frac{p(t)}{\hat{p}}\right)$ captures throughput as a function of pressure at time t relative to the pipeline's rated capacity, reflecting the relationship between pressure and flow volume. The upper bound $\hat{p}$ could be treated as an endogenous variable, determined by design parameters such as pipe material, wall thickness, burial depth, and ambient temperature, which influence a pipeline's structural capacity to safely withstand internal pressure (American Lifelines Alliance, 2001). Incorporating these characteristics would allow the model to reflect differences in physical integrity across firms and enable contracts to adjust bond requirements or refund structures accordingly. Consequently, leak probability would become $q(e, p)$, decreasing in

effort and increasing in pressure, and the agent's utility function would become $U(e, p; c_i, B) = R(p) - c_i e^2 - q(e, p)B - C(p)$, with $C(p)$ denoting pressure-related internal costs. This framework preserves the core mechanism of unobservable safety effort while accounting for the incentive to operate closer to physical limits, emphasizing the importance of bond design in discouraging excessive pressurization without deterring structural investment in safety.

The model could be further enhanced by introducing a formal risk framework that incorporates stochastic variation in leak damages, including rare but catastrophic events. When combined with agent risk aversion and uncertainty in enforcement outcomes, such a structure would enable a richer assessment of contract performance under probabilistic environmental and behavioral conditions. This extension could also allow for conditional forfeiture rules that distinguish between preventable leaks and uncontrollable disasters, such as acts of God or terrorism, where no reasonable level of effort would have avoided the outcome. In such cases, bond design might incorporate narrowly tailored exemptions or insurance-backed protections that apply only to exogenous shocks. To preserve the deterrent function of the bond, these protections must be structured so they do not indemnify negligence or eliminate the operator's financial exposure. Carefully calibrated in this way, insurance mechanisms can enhance fairness without undermining the bond's incentive power.

A related concern involves the risk that a firm will be unable to pay damages following a spill due to insufficient assets. As Shavell (1986) explains, when liability exceeds a firm's net worth, the threat of ex post penalties does not fully deter harmful behavior. This judgment-proof problem weakens the effectiveness of traditional liability rules and creates incentives for firms to underinvest in prevention. In such cases, requiring financial assurance in advance becomes especially important. Reclamation bonds help address this gap by ensuring that resources are

available before any damage occurs, thereby maintaining deterrence even when firms face solvency constraints. Calibrated carefully, the combination of bonding and selective insurance can preserve incentives for precaution while accounting for financial vulnerability and uncontrollable risks.

Additionally, conducting Monte Carlo simulations across key parameters simultaneously for different baseline values would allow the model to examine how optimal contracts adapt under broader uncertainty, including alternative functional forms and relationships between effort and cost, as well as diverse market compositions and operational heterogeneity beyond the single-variable scenarios explored here. Another promising extension involves treating regulatory enforcement as a strategic decision. If monitoring effort is costly, the state may optimally choose when and how frequently to audit firms (Bizzotto & Chiara, 2022). Delegating this responsibility to independent third parties introduces additional complications, such as the potential for collusion or compliance failures, which could justify more stringent or flexible bonding requirements.

Additionally, the information structure itself could be expanded by allowing unobserved cost types to interact with heterogeneous risk preferences or pollution severity, which would influence the configuration and sorting of optimal contracts. Instead of separating firms solely based on observable outcomes, the model could incorporate a menu of bond and refund options tailored to operator-specific characteristics. Additional design innovations, such as group-based contracts, shared liability arrangements, or uncertainty in both enforcement and damage severity, would bring the model closer to institutional realities. Transitioning from discrete cost types to a continuous distribution would also enable finer contract differentiation, better reflecting the diversity of operator technologies, environments, and compliance capabilities (Laffont & Martimort, 2009).

It is also important to recognize that while the principal in this model is assumed to act only to maximize landowner welfare and broader social objectives, real regulatory practice often falls short of this benchmark. Many states depend on substantial revenues from extractive industries, including severance taxes and royalties tied to oil and gas production. These revenues ultimately rely on pipeline infrastructure to move products to market (U.S. Energy Information Administration, 2024). Although pipeline operators themselves usually do not pay severance taxes, this economic linkage can create an institutional tendency to favor continued extraction over strict safety enforcement. As a result, regulatory authorities may enforce bonding requirements less rigorously or delay corrective measures after incidents, which weakens the credibility of financial safeguards meant to protect landowners.

To address such conflicts of interest, some jurisdictions have established independent regulatory commissions that are structurally separated from revenue flows linked to resource extraction. For example, the Canada Energy Regulator mandates that pipeline operators maintain dedicated trust funds to ensure financial capacity for future decommissioning and liability management. This model is designed to enhance public confidence in the integrity of oversight (Canada Energy Regulator, 2024). While government agencies often administer such systems, there is evidence that private risk pooling and industry-led governance structures can also play a meaningful role in managing environmental and safety liabilities. For instance, the Federal Aviation Administration's commercial space launch program incorporates private risk sharing and financial responsibility requirements within a structured public-private framework, illustrating how industry mechanisms can be integrated into effective oversight when accompanied by enforceable standards and transparent financial assurances (Federal Aviation Administration, 2002). Regardless of the

institutional form, sustained public trust depends on the independence of the overseeing body, the sufficiency of financial guarantees, and the credibility of enforcement mechanisms.

In addition to institutional independence, regulators often employ a broader set of financial tools to strengthen accountability. Alongside reclamation bonds, agencies frequently require spill response bonds that secure immediate cleanup resources (Washington State Department of Ecology, n.d.). Some regulators also impose re-bonding requirements, obligating operators to post higher bonds following an incident, since a leak in one section often signals broader structural risks. This is consistent with guidance from the Pipeline and Hazardous Materials Safety Administration, which notes that problems like corrosion or faulty welds are rarely confined to a single location and typically prompt inspections of adjacent segments to identify further vulnerabilities (Pipeline and Hazardous Materials Safety Administration, 2017). In support of this approach, federal authorities have already expanded their enforcement powers to enable more frequent inspections and closer oversight after pipeline failures (U.S. Department of Transportation, 2010).

Mathematically, such a re-bonding mechanism could be incorporated into the model by allowing the bond amount B_i offered to each firm type $i \in \{L, H\}$ to vary based on the firm's observed history. For instance, if a leak is recorded in a previous period, the updated bond in the following round could be defined as $B_i' = B_i + \Delta_i$, where $\Delta_i > 0$ is a penalty increment specific to type. More generally, bond amounts can be written as $B_i(h)$, where $h \in \{0, 1\}$ is a history flag indicating whether the firm had a recent safety violation ($h = 1$) or not ($h = 0$). This approach preserves the single-period logic of the baseline model while introducing a simple mechanism to condition future bond requirements on past behavior. By anticipating higher financial

requirements after an incident, firms have an added incentive to exert effort early, even without expanded monitoring or enforcement.

To maintain both regulatory solvency and operator viability, some bond structures allow funds to accumulate interest in dedicated trust accounts, with these earnings used to support additional oversight or to replenish reserves without placing sudden capital burdens on firms. Bond schedules that decline over time in the absence of new incidents can also reward consistent safety performance and ease financial strain on compliant operators. These mechanisms help protect smaller companies from insolvency after a single adverse event and reduce the likelihood that the state will confront unfunded cleanup obligations (WyoFile, 2011).

While this study focuses only on a static menu of reclamation bonds for simplicity, the broader regulatory experience shows that carefully designed financial instruments, whether through independent oversight bodies, interest-earning trusts, or flexible re-bonding requirements, play an essential role in complementing the incentive structures explored in this model. They help ensure that pipeline safety is maintained even where direct monitoring is limited or institutional incentives are misaligned. Beyond these static mechanisms, environmental management theory underscores the value of dynamic regulatory strategies. Holling (1978) characterizes adaptive management as an iterative process in which policies function as experiments, generating feedback that informs subsequent revisions. Rather than relying on fixed designs, this approach promotes continuous learning through systematic observation and adjustment. Applied to pipeline oversight, it supports a framework in which bond requirements and inspection protocols are modified over time in response to safety records and operational outcomes. In practice, this type of layered strategy is evident where the BLM (n.d.) requires applicants to provide upfront cost recovery payments that

fund environmental reviews under the National Environmental Policy Act, including cases where an Environmental Impact Statement is necessary (Code of Federal Regulations, 2025).

For instance, regulatory instruments can also evolve in response to physical deterioration over time. As pipelines age, the probability of failure increases even when initial safety standards are met. A natural extension of this model would capture this dynamic by allowing the probability of a leak to depend not only on effort but also on time, t . One such formulation could be $q(e, t) = q_0 e^{-\alpha e + \beta t}$, where $\beta > 0$ reflects the rate at which underlying risk accumulates as infrastructure degrades.

This structure preserves the existing deterrent role of effort while introducing the empirically supported fact that aging pipelines face rising hazard rates due to corrosion, pressure cycles, and fatigue. This relationship is well established in reliability engineering, where failure probability is commonly modeled using time-dependent functions such as Weibull or exponential forms (Rausand and Høyland, 2003). This time-dependent risk structure also influences the firm's effort decision when combined with discounting and firm heterogeneity. If future bond forfeiture is heavily discounted and the probability of a leak grows rapidly over time, the present benefit of investing in safety can fall below its cost. Under these conditions, even rational firms may choose to forgo preventive effort. Research in reliability engineering confirms that optimal maintenance or mitigation effort can reach zero when the discounted cost of action outweighs the expected benefit (Van der Weide, Pandey, & van Noortwijk, 2010). This outcome highlights the importance of bond designs that front-load incentives or adjust timing to ensure continued investment in prevention. Such measures extend beyond traditional bonding, ensuring that regulatory processes are adequately resourced while also adapting financial obligations to project complexity. Collectively, these refinements expand the model's relevance across diverse regulatory and informational contexts.

Conclusion

This study demonstrates that well-designed reclamation bonds transform financial guarantees into self-enforcing instruments that address hidden costs and unobservable effort, motivating operators to invest in safety and reducing expected harm to landowners even where traditional monitoring fails. This is especially pressing given the fragmented enforcement landscape and the costly consequences of spills that are too frequently absorbed by communities and landowners. Simulations grounded in realistic regulatory and industry data also confirm that a carefully structured menu of bond amounts leads both low-cost and high-cost operators to take substantial preventive action, minimizing the chances of environmental damage. Moreover, the model shows that this contract approach is robust as it maintains incentive compatibility and ensures operator participation even as economic conditions, environmental stakes, or the makeup of the operator pool change.

At the same time, this work underscores the need to strengthen these tools further. Introducing re-bonding provisions that raise collateral after an incident would ensure that financial guarantees keep pace with revealed risks. Building adaptive elements into bond design could allow regulators to tighten or relax requirements as they learn more about operator behavior and safety records over time. Extending the model to capture operator risk aversion, random catastrophic losses, or operational pressures would also provide a more complete view of the tradeoffs at play. Moving from discrete types to continuous cost differences would enable even finer contract tailoring that better reflects the diversity of pipelines and operating environments.

In short, this study demonstrates that reclamation bonds, grounded in both contract theory and institutional practice, can serve as a viable tool for aligning firm behavior with policy objectives.

In regulatory environments where incomplete information and fragmented oversight hinder effective governance, these financial instruments help promote preventive investment by firms while also offering a layer of protection to landowners who face the localized consequences of environmental harm. Although the objectives of regulators and landowners may differ, with the former focused on systemic risk reduction and the latter seeking compensation and security, the bond mechanism provides a workable structure for managing these intersecting interests. While the present model adopts an instrumental perspective centered on encouraging precaution, future research may examine whether such strategies are consistent with broader welfare outcomes, particularly in cases where investments in safety do not necessarily maximize aggregate wealth.

Acknowledgements

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Appendix

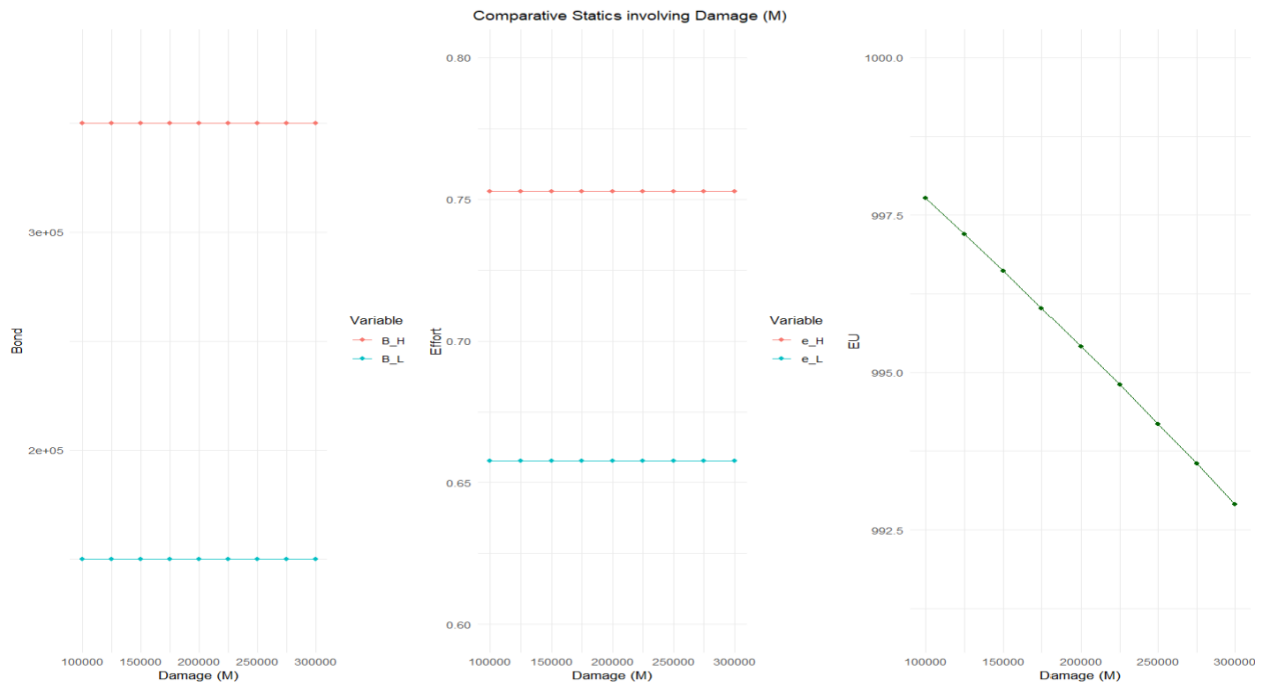
Table 1: Summary of baseline model parameters

Parameters	Description	Value
c_L	Marginal effort cost (low type)	1,000
c_H	Marginal effort cost (high type)	2,000
M	Environmental damage from a leak	200,000
λ	Landowner wealth	1,000,000
q_0	Baseline leak probability	0.05
α	Leak decay to effort parameter	0.2
γ	Share of low-cost operators	0.5
R	Pipeline operator revenue	18,000,000

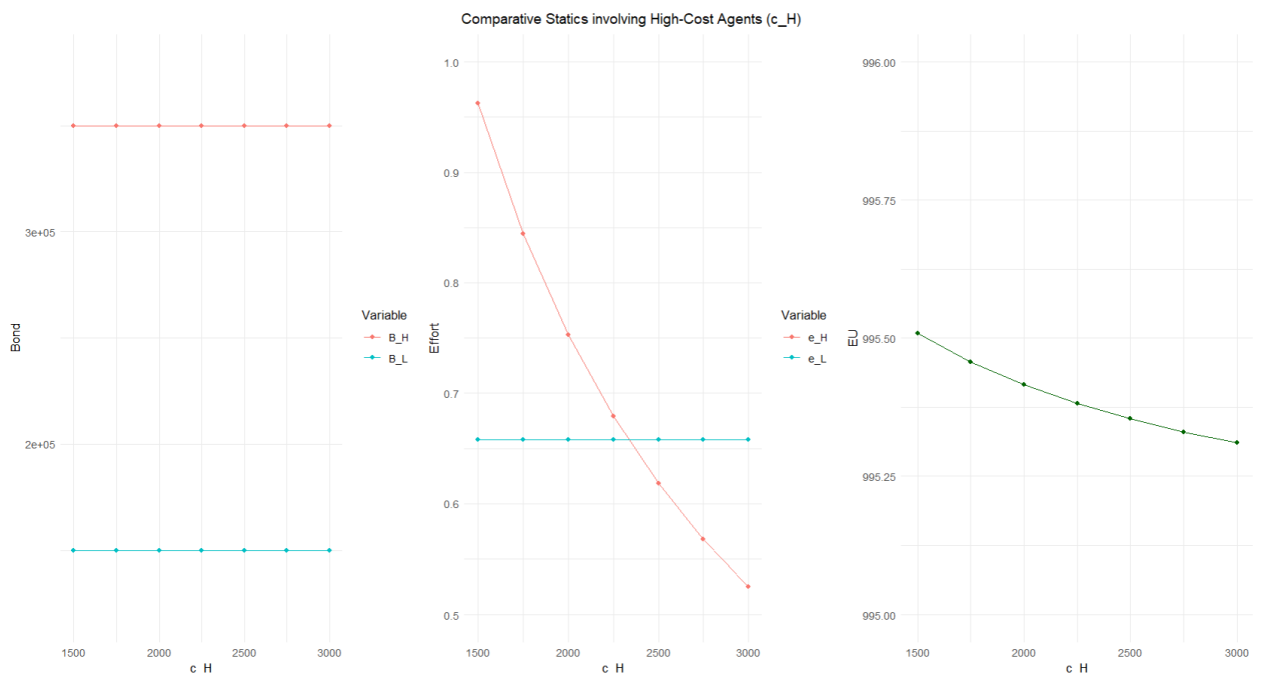
Table 2: Summary of comparative statics results

Parameters	Range Tested	Effect on Bond Levels	Effect on Effort Levels	Effect on Expected Utility
M	100,000 to 300,000	No change	No change	EU falls as M increases
c_H	1,500 to 3,000	No change	c_H falls as c_H increases	EU falls as c_H increases
c_L	750 to 2,500	No change	c_L falls as c_L increases	EU falls as c_L increases
R	16,000,000 to 20,000,000	No change	No change	No change
λ	800,000 to 1,200,000	No change	No change	EU increases as λ rises
γ	0.1 to 0.9	No change	No change	EU falls as γ increases
α	0.05 to 0.30	No change	Effort levels rise	EU increases as α rises
q_0	0.01 to 0.10	No change	Effort levels rise	EU falls as q_0 increases

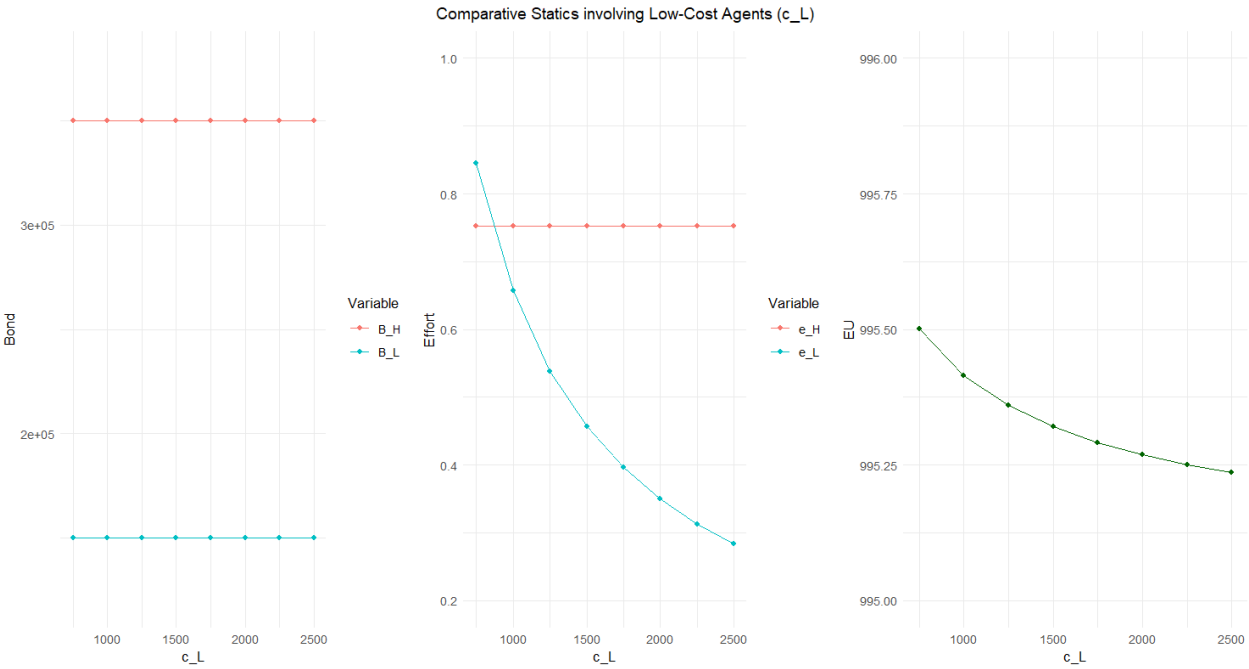
Plot 1: Effect of Damage (M) on Contract Parameters



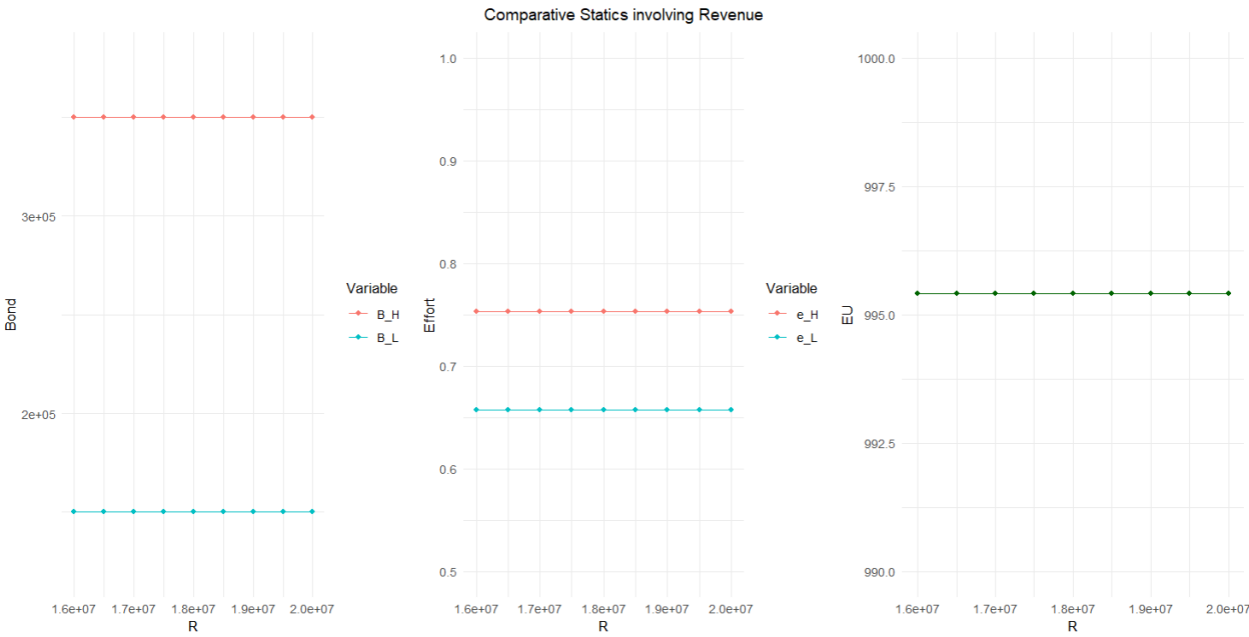
Plot 2: Effect of High-cost Agents' Effort Cost (C_H) on Contract Parameters



Plot 3: Effect of Low-cost Agents' Effort Cost (c_L) on Contract Parameters



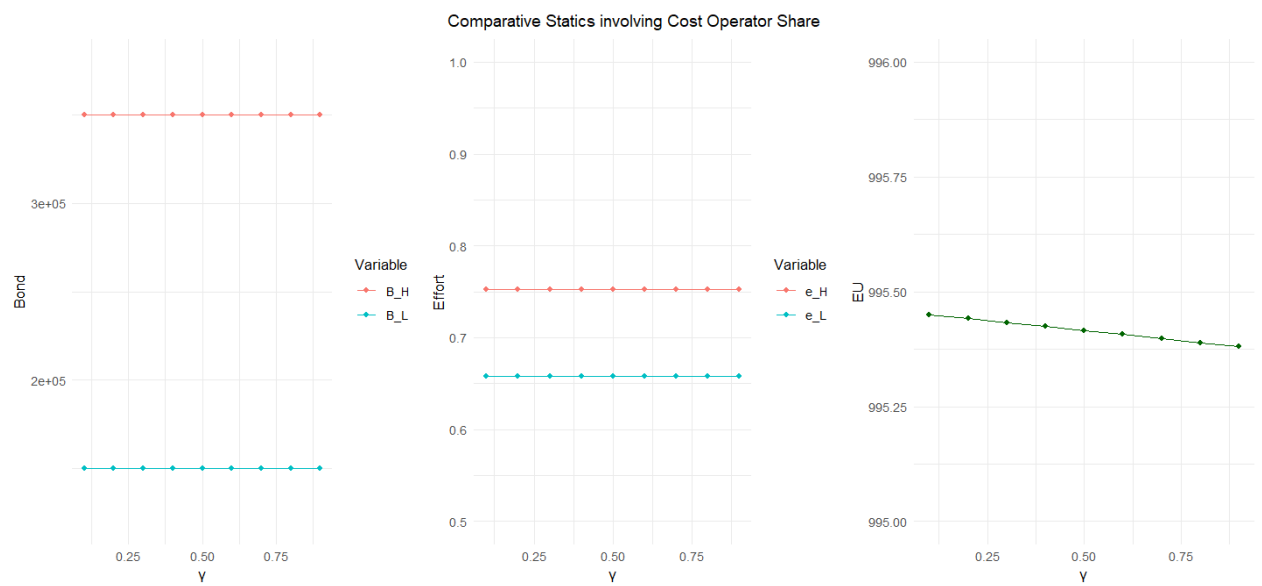
Plot 4: Effect of Agent's Revenue (R) on Contract Parameters



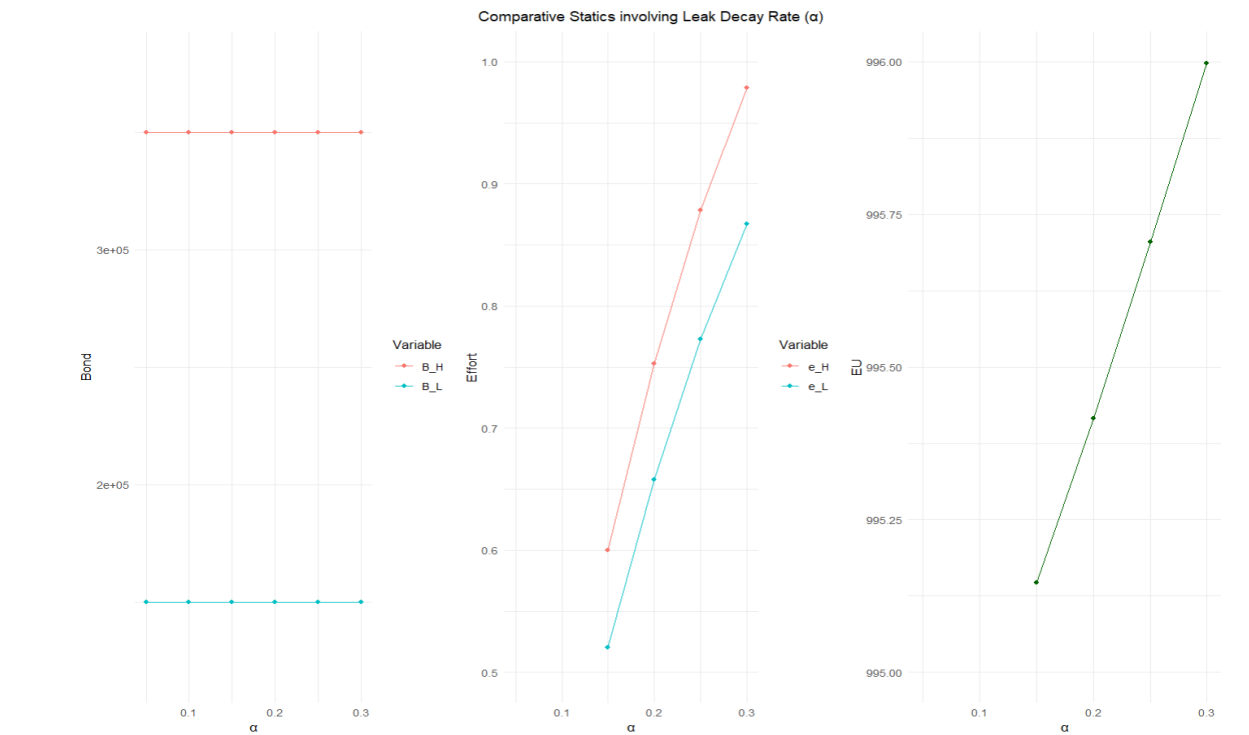
Plot 5: Effect of Landowner Wealth (λ) on Contract Parameters



Plot 6: Effect of Cost Operator Share (γ) on Contract Parameters



Plot 7: Effect of Effort Decay Rate on Contract Parameters



Plot 8: Effect of Baseline Leak Probability on Contract Parameters

